

Dynamic Web Applications

IS53076A

Lecture 1.2
Module Content

Contents

- Why this module?
- What concepts will we be learning?
- What technology stack will we be using?
- What happens when we browse to a web page?

Why this module is important

- These skills are in demand!
 - That means jobs!
- The underlying technologies (data and web technologies) are critical components of modern computing
- You can make fun stuff with this knowledge!

Why SQL (Structured Query Language)?

- Still widely used, and likely to be for a long time
- NoSQL databases often use 'SQL-like' terminology
- It is fundamental language for dealing with data

Description	Rank 6 Months to 10 Sep 2025	Rank YoY Change	Median Salary	Median Salary YoY Change	Historical Absolute & Relative Jobs Vacancies	Live Jobs
SQL	12	▼ -1	£57,500	-2.74%	5,013 10.09%	3,624

Description	Rank 6 Months to 10 Sep 2025	Rank YoY Change	Median Salary	Median Salary YoY Change	Historical Absolute & Relative Jobs Vacancies	Live Jobs
Social Skills	1	↔ 0	£47,500	-9.52%	13,990 28.16%	11,481
Finance	2	↔ 0	£65,000	-7.14%	11,656 23.47%	7,932
Degree	3	▲ +5	£53,065	-3.51%	9,641 19.41%	6,389
Problem-Solving	4	▲ +2	£45,000	-12.62%	9,385 10.89%	6,301
Agile	5	▼ -2	£67,500	+3.10%	8,096 16.30%	4,982
Microsoft	6	▼ -1	£46,500	-7.57%	7,825 15.75%	5,330
Azure	7	↔ 0	£60,000	-4.00%	7,315 14.73%	3,997
Senior	8	▼ -4	£67,000	+3.07%	6,142 12.36%	3,924
Analytical Skills	9	▲ +4	£52,500	-	6,082 12.24%	4,456
Artificial Intelligence	10	▲ +30	£70,000	-6.66%	5,353 10.78%	3,413
Python	11	▲ +4	£70,000	-3.33%	5,208 10.48%	4,245
SQL	12	▼ -1	£57,500	-2.74%	5,013 10.09%	3,624
AWS	13	▼ -1	£75,000	-	4,797 9.66%	2,935
Continuous Improvement	14	▲ +15	£56,000	-6.66%	4,693 9.45%	3,217
Windows	15	▲ +3	£38,325	-14.83%	4,612 9.28%	1,652
Mentoring	16	▲ +4	£64,500	+2.15%	4,559 9.18%	3,516
Marketing	17	▲ +16	£45,000	-10.00%	4,251 8.56%	3,213
Developer	18	▼ -9	£62,500	+4.16%	4,229 8.51%	2,075
Analyst	19	▼ -5	£44,500	-1.11%	4,169 8.39%	3,067
Inclusion and Diversity	20	▲ +5	£55,000	-4.34%	4,144 8.34%	1,619
Analytics	21	▲ +14	£60,000	-7.69%	3,930 7.91%	3,674
Customer Service	22	▲ +8	£32,000	-11.11%	3,920 7.89%	2,416
DevOps	23	▼ -2	£70,000	+3.70%	3,882 7.82%	1,924
Lead	24	▼ -8	£75,000	+7.14%	3,777 7.60%	2,104
Software Engineering	25	▼ -15	£75,000	+15.38%	3,676 7.40%	2,151
Security Cleared	26	▼ -2	£62,500	+4.16%	3,467 6.98%	2,939
Law	27	▲ +19	£55,000	+10.00%	3,370 6.78%	1,308
Project Management	28	▼ -5	£60,000	+9.09%	3,268 6.58%	3,404
Retail	29	▲ +10	£42,500	-15.00%	3,220 6.48%	2,342
CI/CD	30	▲ +4	£72,500	+3.57%	3,205 6.45%	1,650

Data science / data engineering

- Relational databases and SQL are key technologies in data science and data engineering

Some jobs this module could help towards...

- Front-end web developer
- Back-end web developer
- Full-stack developer
- Mobile app developer
- API developer / integration engineer
- Web performance optimization
- A variety of database and data roles
 - Database developer
 - Database manager
 - Data scientist
 - Data engineer
 - SQL DBA (DataBase Administrator)
 - SQL Analyst
- Cybersecurity roles
- Also lots of scope for freelance work
 - Web development
 - CMS developer (WordPress, etc)
 - etc

Questions we will answer

- What happens behind the scenes when I browse the web?
- What are the key technologies and concepts that hold the web together?
- What exactly is in the cloud and what goes on there?
- How can I create my own website?
- How can I make the website dynamic and interactive?
- How can I make my website public?
- How do websites store my data?
- How can I make sure my data integrity is maintained?
- How do I keep data secure?

Key data concepts we will learn

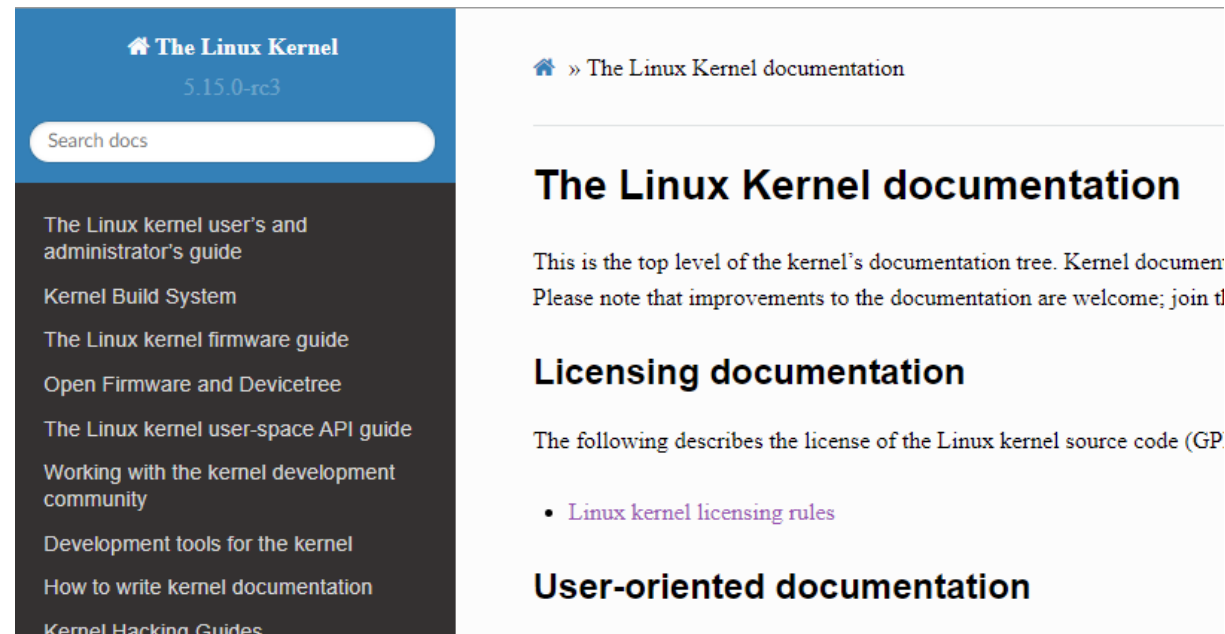
- Relational databases
- SQL
- Entity Relationship Diagrams
- Primary and foreign keys
- CRUD

Key web concepts we will learn

- Servers and clients
- HTTP
- Web sites and web apps
- Dynamic and static sites
- Tiered architectures
- HTML
- Javascript
- Hosting
- APIs

What happens when we browse the web

- Point your browser here:
 - <https://www.kernel.org/doc/html/latest/>
- We see the Linux documentation:



What happened between you clicking on the link
and seeing the page?

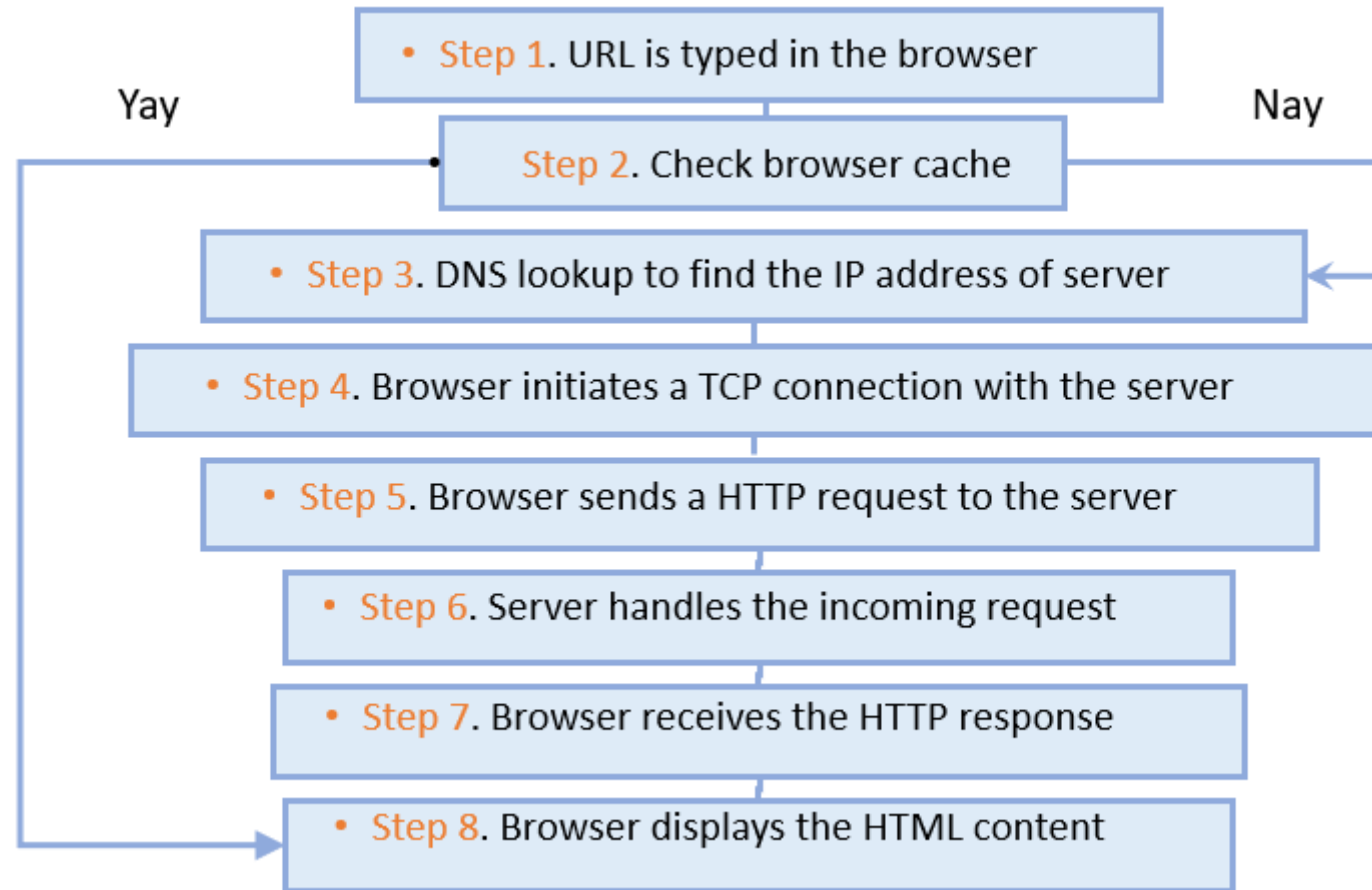
What happens when we browse the web

- You probably intuitively understand that there are a few components:
 - The computer you are on and the browser you are running
 - Some computer on the internet that is running the website
 - Some data (i.e. the website contents)
 - Some way for the computers to talk to each other

But how do these components come together to allow you to browse the site?

The steps in a web request

- Here's what happens:



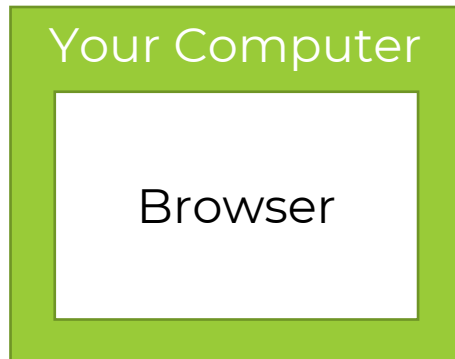
That's a lot of new stuff!

- **Server:** The web server, which is running the web site or web application
- **DNS:** Domain Name Server, are the Internet's equivalent of a phone book. They maintain a directory of domain names and translate them to Internet Protocol (IP) addresses.
- **IP:** Internet Protocol, identifies a machine connected to a network.
- **TCP:** Transmission Control Protocol. Facilitates communication between computers.
- **HTTP:** Hyper-text Transfer Protocol. Main protocol used to transfer hypertext pages (HTML) from a web server to a browser.

We won't worry about the others in this module. We'll just focus on HTTP.

The steps in a web request

We have 3 computers running 3 bits of software

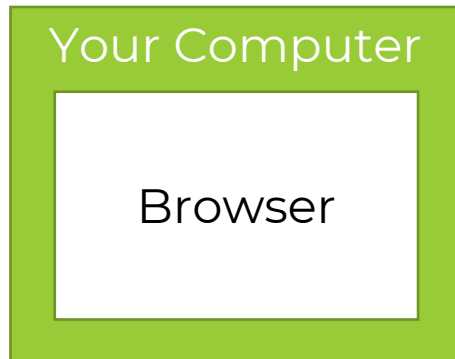


The steps in a web request

You start your browser and go to the website

Step 1. URL is typed in the browser

Step 2. Check browser cache. If page is there display it

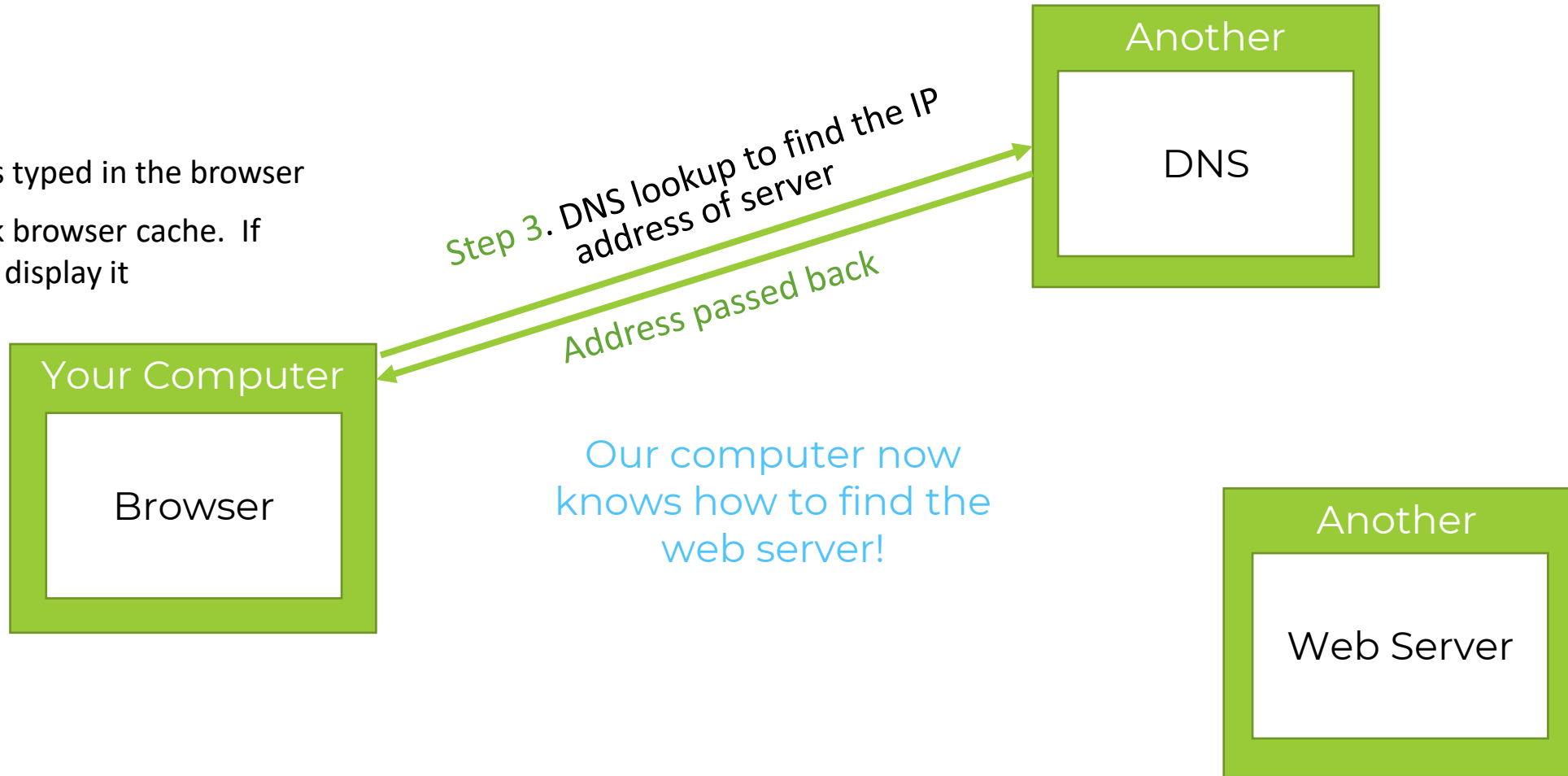


The steps in a web request

The DNS lookup is performed

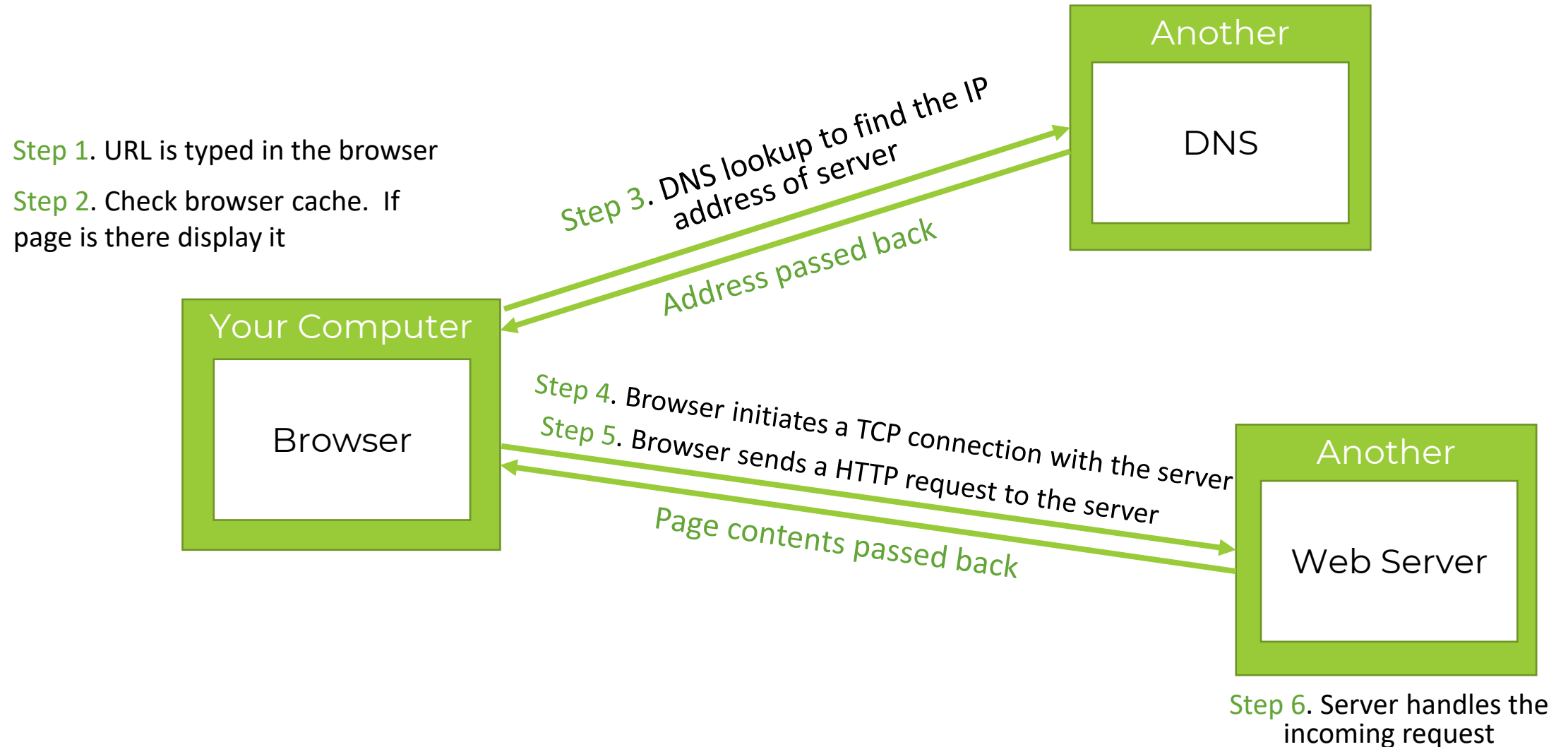
Step 1. URL is typed in the browser

Step 2. Check browser cache. If page is there display it



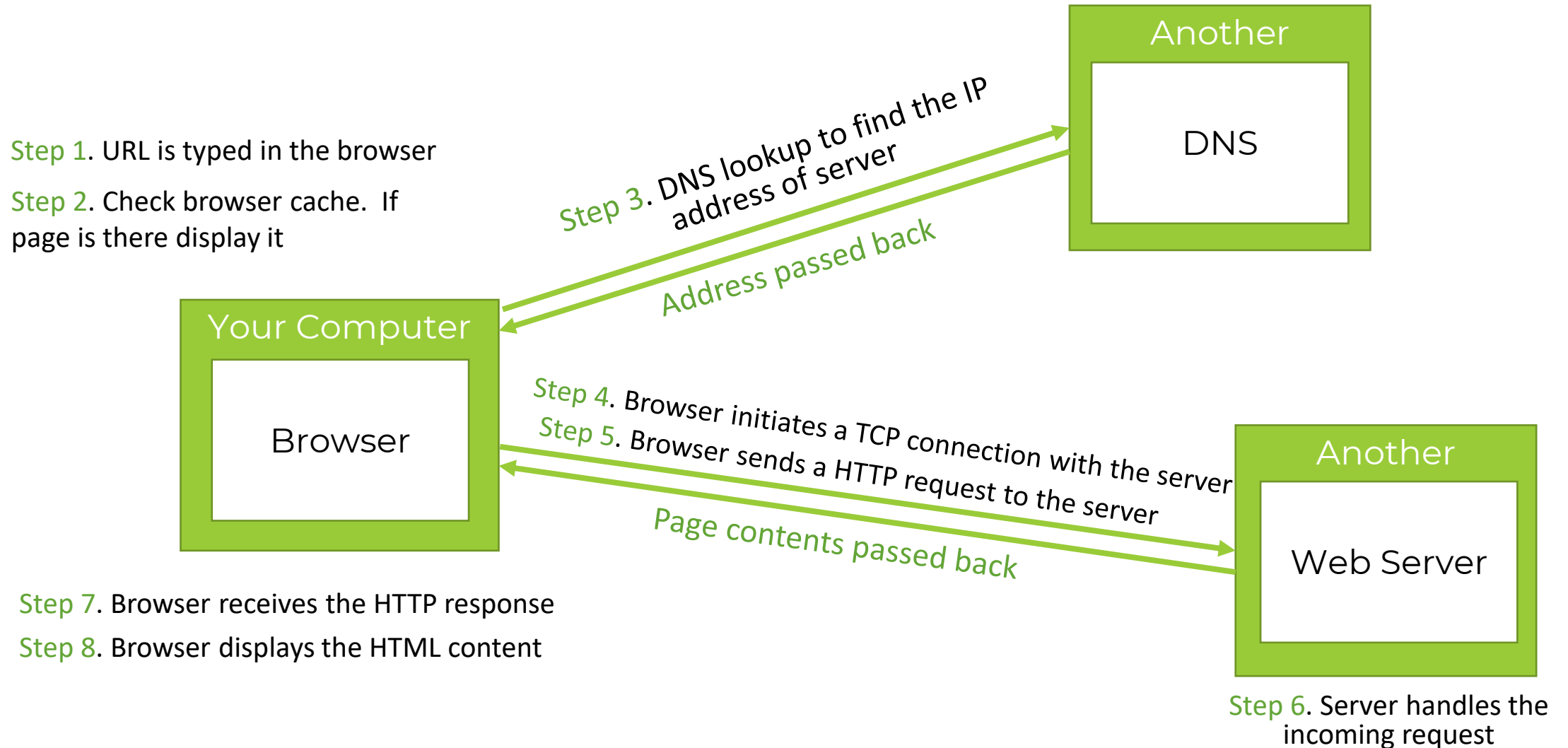
The steps in a web request

Our computer requests the web page contents



The steps in a web request

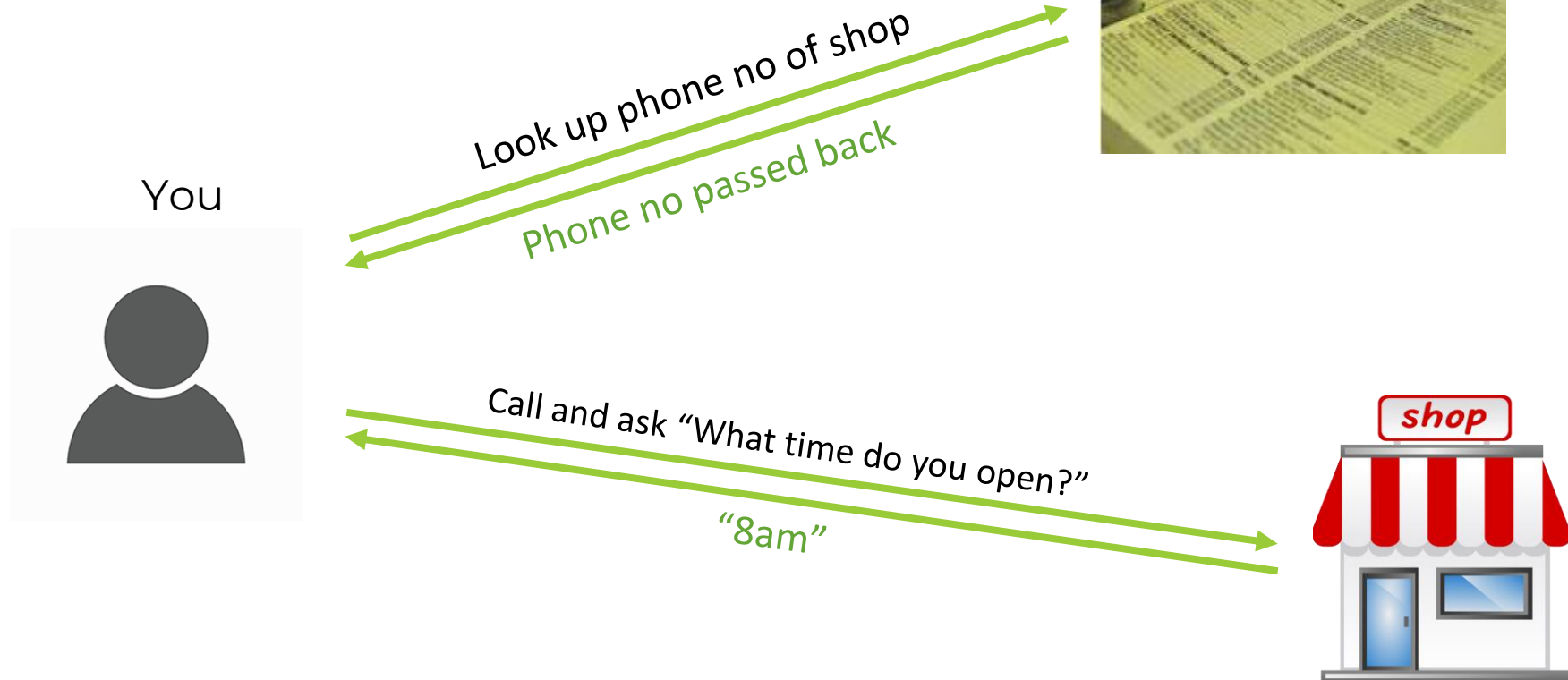
The browser displays the page



The steps in a web request

Just think of it as like a phone directory lookup to get a phone number

Telephone Directory

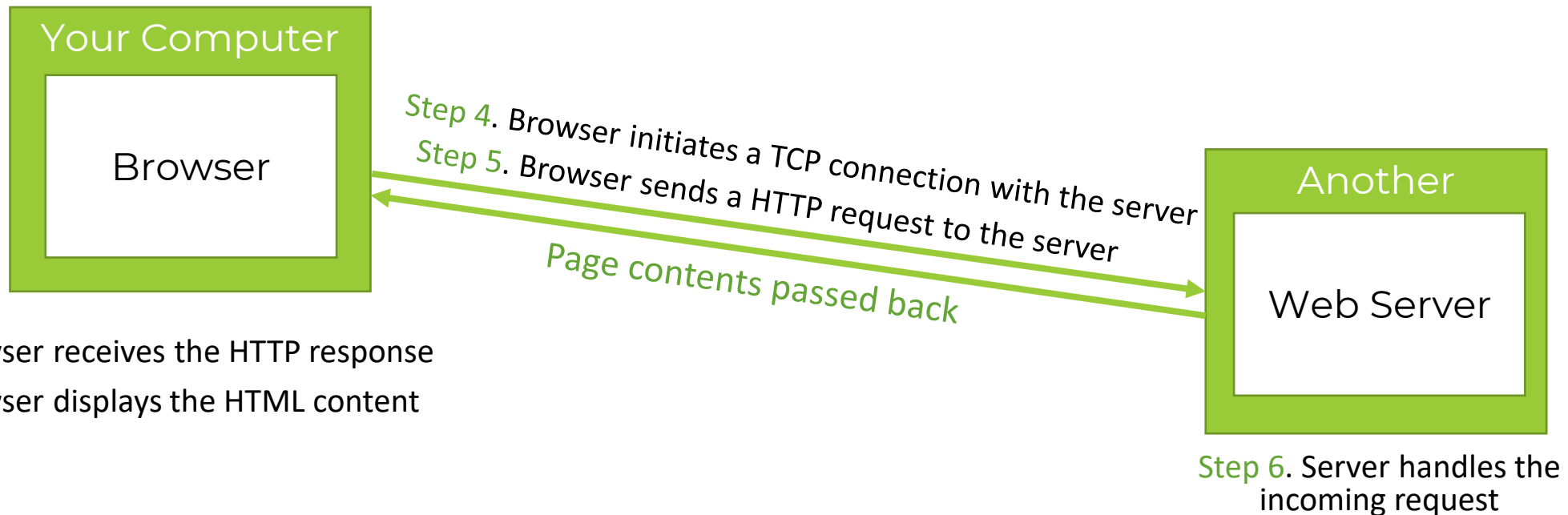


The steps in a web request

For the purposes of this module, don't worry about the DNS. Just know that it's there. Let's focus on these components and steps:

Step 1. URL is typed in the browser

Step 2. Check browser cache. If page is there display it

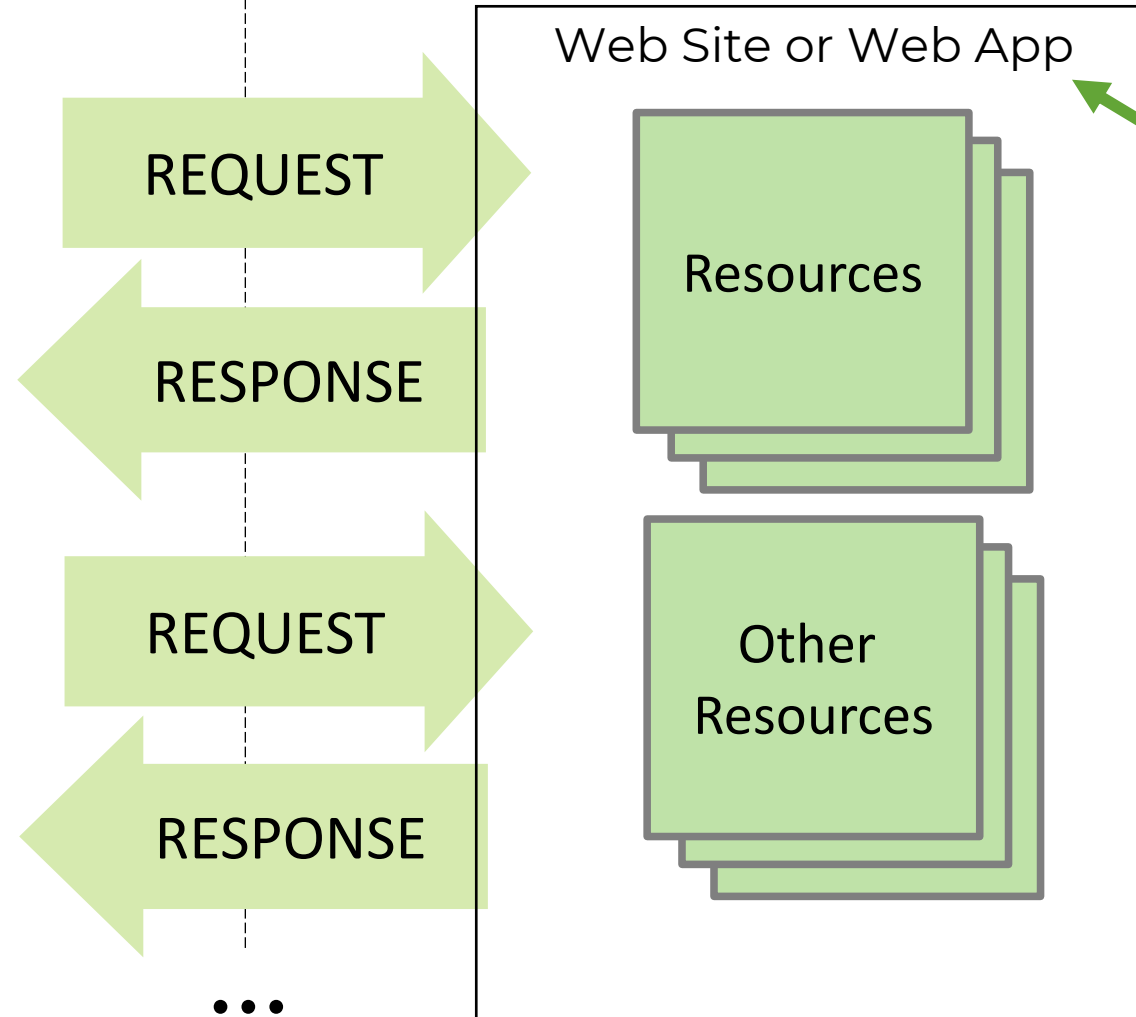


HTTP (Hyper Text Transfer Protocol)

Examining the HTTP request in more detail

Client

Web Server



It's a conversation between the client, running your browser, and the web server, running your web site or web app.

Understanding how this conversation works is the key thing for this module!

Building a few of these is what you will do in the labs

Technology stacks: web

- There are lots of programming languages we can use to build web sites and web apps

Python PHP Javascript
Ruby C# CSS HTML
Java

These are just some of
the main ones!

- Layered on top of these there are numerous frameworks

Express Angular
Django Vue
Spring Rails
Ember Backbone React

These are just some of
the main ones!

Technology stacks: data

- There are lots of ways we can build data stores



Article [Talk](#)

List of relational database management systems

From Wikipedia, the free encyclopedia

List of software [\[edit \]](#)

- | | | | | | |
|--|--|--|--|---|---|
| <ul style="list-style-type: none">• 4th Dimension• Adabas D• Airtable• Apache Derby• Aster Data• Amazon Aurora• Altibase• CA Datacom• CA IDMS• Clarion• ClickHouse• Clustrix• CockroachDB• CSQL• CUBRID• DataEase• DataFlex• Database Management Library• Dataphor | <ul style="list-style-type: none">• dBase• Derby (aka Java DB)• Empress Embedded Database• Exasol• EnterpriseDB• eXtremeDB• FileMaker Pro• Firebird• FrontBase• Google Fusion Tables• Greenplum• GroveSite• H2• Helix• HSQLDB• IBM Business System 12 (historical)• IBM DB2• IBM Lotus Approach | <ul style="list-style-type: none">• IBM DB2 Express-C• Infobright• Informix• Ingres• InterBase• InterSystems Caché• InterSystems IRIS Data Platform• LibreOffice Base• Linter• MariaDB• MaxDB• Microsoft Access• Microsoft Jet Database Engine (part of Microsoft Access)• Microsoft SQL Server• Microsoft SQL Server Express• SQL Azure (Cloud SQL Server)• Microsoft Visual FoxPro• Mimer SQL | <ul style="list-style-type: none">• MonetDB• mSQL• MySQL• Nettezza• NexusDB• NonStop SQL• NuoDB• Omnis Studio• OpenLink Virtuoso (Open Source Edition)• OpenLink Virtuoso Universal Server• OpenOffice.org Base• Oracle• Oracle Rdb for OpenVMS• Panorama• Paradox• Percona Server for MySQL• Percona XtraDB Cluster | <ul style="list-style-type: none">• Pervasive PSQL• Polyhedra• PostgreSQL• Postgres Plus Advanced Server• Progress Software• Raima Database Manager (RDM)• RDM Server• R:Base• RethinkDB• SAND CDBMS• SAP HANA• SAP Adaptive Server Enterprise• SAP IQ (formerly known as Sybase IQ)• SingleStore• Snowflake Cloud Data Warehouse• solidDB• SQL Anywhere (formerly known as Sybase Adaptive Server Anywhere and Watcom SQL) | <ul style="list-style-type: none">• SQLBase• SQLite• SQream DB• SAP Advantage Database Server (formerly known as Sybase Advantage Database Server)• Teradata• Tibero• TiDB• TimesTen• Trafodion• Unisys RDMS 2200• UniData• UniVerse• Vectorwise• Vertica• VoltDB |
|--|--|--|--|---|---|

How not to get lost in the sea of technologies

- Take one popular technology stack as your learning platform
- Learn it well
- The concepts are generally transferrable

Main technology stack for this module

- HTML / CSS For describing our content
- Javascript For dynamic behaviour in the browser and web app
- Node.js To run the javascript web app
- Express To provide a framework for our web app
- MySQL To store our data

Labs environment

- We will be using virtual servers provided by Goldsmiths to host your web applications
- You will create virtual servers, running Linux
- You will build your web apps on your computer, push them to Github and pull them onto your virtual server
- You will need to use terminal sessions...
- ...and learn a few Linux commands

```
Welcome to Ubuntu 20.04.3 LTS (GNU/Linux 5.4.0-91-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/advantage
```

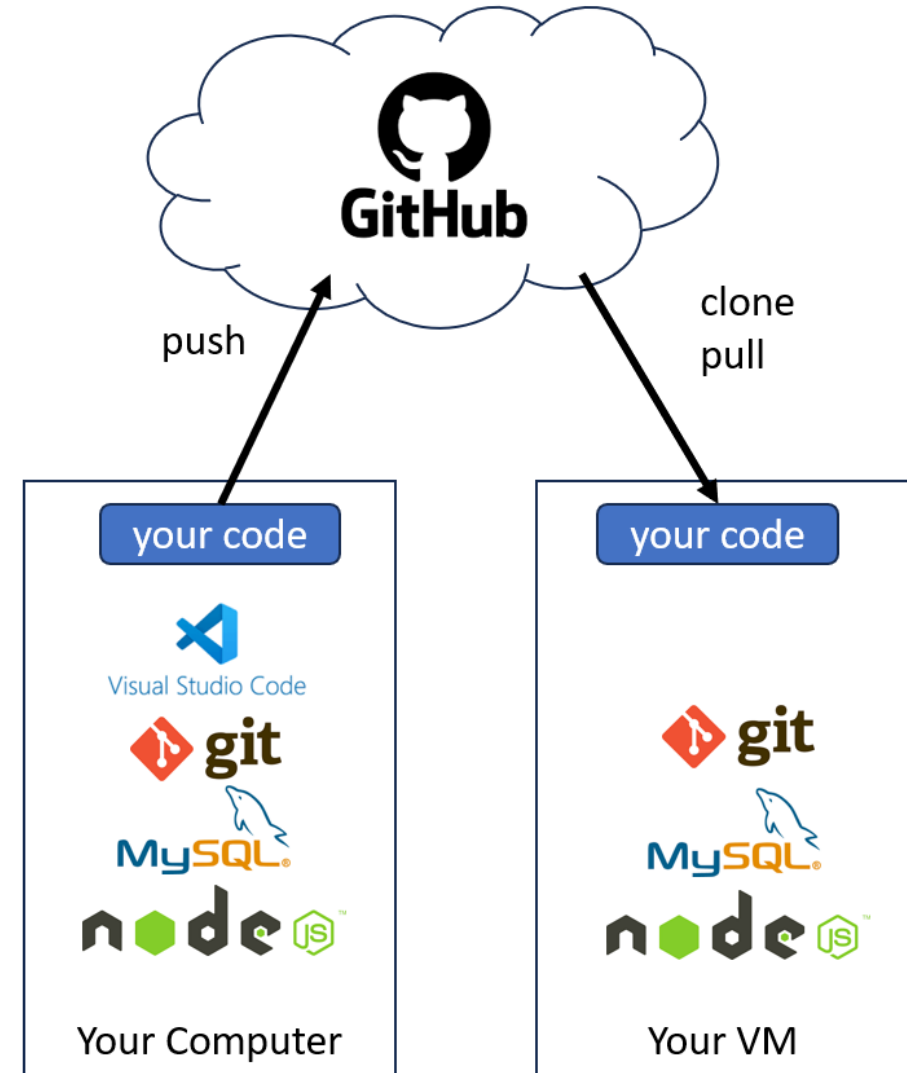
Lab projects

You will create and submit 4 web applications, increasing in complexity:

- MyPage
- Express
- Thirsty Student
- Berties Books

Development Process

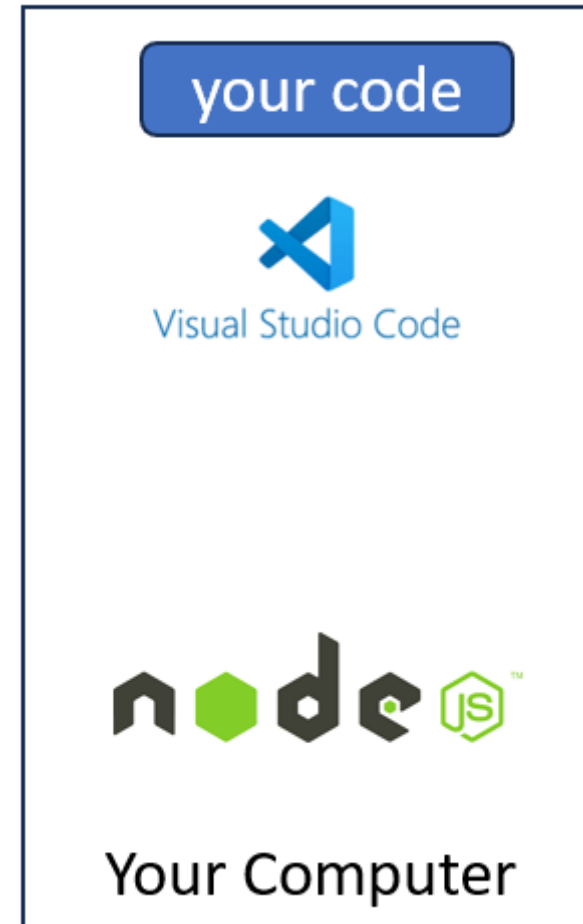
The aim is to eventually end up with an environment like this and an understanding of the process of developing and deploying web applications.



First Lab

In the first lab you will be setting up these parts of your development environment.

You will also build a very simple web application.



END OF
SLIDES